Consider the following sets of simultaneous equations.

**Set 1**

Solve

$$4x_1 + 2x_2 = -3 \tag{1}$$
$$2x_1 + x_2 = 2 \tag{2}$$

for $x_1$ and $x_2$.

**Set 2**

Solve

$$4x_1 + 2x_2 = -3 \tag{3}$$
$$2x_1 - x_2 = 2 \tag{4}$$

for $x_1$ and $x_2$.

**Classifying Matrices**

In the following, identify:

1. the matrices that are likely to have a unique solution to $Ax = b$
2. the matrices that represent systems of equations that can be solved using the Jacobi method
3. the matrices that represent systems of equations that can be solved using $LU$ factorization (after rearranging the rows if necessary)

$$B_1 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 2 \\ 0 & -2 & 1 \end{bmatrix}$$

$$B_3 = \begin{bmatrix} 2 & 1 & -1 & 3 \\ 0 & 3 & 1 & 4 \\ 4 & 2 & -2 & 6 \\ 1 & 0 & 2 & -1 \end{bmatrix}, \quad B_4 = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & -1 & 4 & 1 \\ 1 & 1 & 2 & 1 \\ 2 & 0 & 2 & 3 \end{bmatrix}, \quad B_5 = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \end{bmatrix}$$

For each of the following systems of equations,

1. Solve the equations for $x$ and $y$.
2. Find the determinant of the associated matrix
3. Calculate the condition number.

> **Tip**
>
> Note that the matrix
>
> $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
>
> has determinant $|ad - bc|$ and its inverse is
>
> $$\frac{1}{|ad - bc|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

**Set 1**

$$2x + y = 3 \tag{5}$$
$$2x + 1.001y = 0 \tag{6}$$

**Set 2**

$$2x + y = 3 \tag{7}$$
$$2x + 1.002y = 0 \tag{8}$$

**Set 3**

$$2x + y = 3 \tag{9}$$
$$2x + 1.001y = -0.001 \tag{10}$$